SUPPORTING ADULT GUIDANCE

Cyclic quadrilaterals

Circle Theorems · Lesson 5. For any supporting adult — teacher, practitioner, parent or carer.
You do not need to be a mathematics specialist to support this lesson.

What this lesson does

This lesson establishes that opposite angles of a cyclic quadrilateral are supplementary (sum to 180°) — the fourth angle-in-a-circle theorem and the mid-unit checkpoint. Learners discover it by dragging all four vertices of a quadrilateral inscribed in a circle and watching both opposite-pair sums hold at 180°, then prove it from the angle at the centre (the two relevant central angles fill 360°). The exterior-angle corollary is included as extension.

Driving Question: *Why do opposite corners of a circle-quad always add to 180°?*

The mathematics, in plain terms

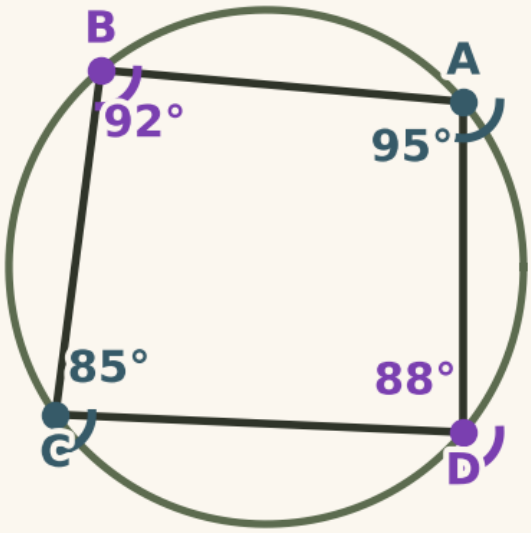
If ABCD is cyclic then ∠A + ∠C = ∠B + ∠D = 180°. Proof: ∠A and ∠C are inscribed angles standing on the two arcs cut by diagonal BD; the corresponding central angles are 2∠A and 2∠C and together they complete a 360° revolution, so 2∠A + 2∠C = 360° and ∠A + ∠C = 180°. Consequently the exterior angle at any vertex equals the interior angle at the opposite vertex.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Find the opposite angle

ABCD is a cyclic quadrilateral. Reading its angles, ∠A = 95° and ∠B = 92°. Find ∠C and ∠D.



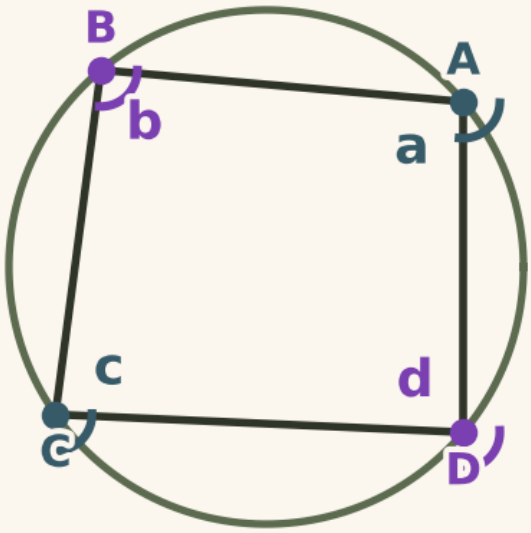
1. Opposite angles add to 180°: ∠A + ∠C = 180°. ∠C = 180 – 95

2. And ∠B + ∠D = 180°. ∠D = 180 – 92

∠C = 85° and ∠D = 88°, because opposite angles of a cyclic quadrilateral add to 180°.

Example 2 — Why they add to 180°

Prove ∠A + ∠C = 180° for a cyclic quadrilateral. Let ∠A = a and ∠C = c.



1. ∠A stands on arc BCD; ∠C stands on arc BAD (the two arcs of diagonal BD).	two arcs
2. Their central angles are 2a and 2c (angle at the centre).	2a and 2c
3. These two central angles make a full turn: 2a + 2c = 360°.	full 360° turn

Dividing by 2: a + c = 180°, i.e. ∠A + ∠C = 180°. Opposite angles of a cyclic quadrilateral are supplementary. ■

Discover the fixed sum, then prove it from the keystone

Let the learner reshape the quadrilateral and watch both opposite-pair sums stay at 180° before stating the rule. Present the proof, as with the previous two theorems, as the keystone applied to a full turn — reinforcing the single root idea. Stress OPPOSITE (not adjacent) angles, since confusing the two is the common error. Use this checkpoint to review all four angle theorems together before the unit turns to tangents and chords; introduce the exterior-angle corollary once the main result is secure.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	See both opposite pairs of a circle-quad add to 180°.	~9 min
Movement	Find an opposite angle: 180° minus the given, with reasons.	~11 min
Reflection	Circle Theorem Lab: drag all four vertices, sums stay 180°.	~12 min
Creativity	Prove it: two central angles fill a 360° turn.	~9 min
Rest	Mid-unit checkpoint — the four angle theorems in hand.	~2 min
Nutrition	Intellectual: four theorems closing into one whole.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Adding adjacent angles instead of opposite ones.
'Which angle is OPPOSITE — across the shape, not next to it? Opposite angles add to 180°.'

Using the rule when the vertices are not all on a circle.
'Are all four corners on the circle? Only then is it cyclic and the rule applies.'

Giving the value with no reason.
'Correct — and the reason is "opposite angles of a cyclic quadrilateral add to 180°".'

Thinking all four angles are equal or add to 180° in any pairing.
'Only the two OPPOSITE pairs each add to 180°. Adjacent angles generally do not.'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the angle at the centre theorem (Lesson 2), that angles at a point add to 360°, and that the angles of a quadrilateral add to 360°. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner adds two adjacent angles, or gives a value with no reason. The aim is a question that fixes the opposite pairing and the reason.

- "Which angle is directly opposite the one you know — across the quadrilateral?"
- "Opposite angles add to 180°, so how do you find the missing one?"
- "What is your reason? Name the theorem."
- "Are all four vertices really on the circle?"

If they stall, that is the moment for a Reconnection Route or the lesson's prompt and hint — not for the answer.

Questions you might have

Why do opposite angles add to 180°?

Each opposite angle is half of a central angle, and the two central angles (on either side of a diagonal) make a full 360° turn. Half of 360° is 180°, so the two opposite angles add to 180°.

Does it work for adjacent angles?

No — only opposite angles are supplementary. Adjacent angles of a cyclic quadrilateral do not generally add to 180°.

What is the exterior-angle result?

Because interior opposite angles add to 180°, and an interior angle and its own exterior angle also add to 180° (straight line), the exterior angle at a vertex equals the interior angle at the opposite vertex.

How does this fit with the other theorems?

It is the fourth angle-in-a-circle theorem, and like the semicircle and same-segment results it follows from the angle at the centre. Together they form a complete little family — the point of the mid-unit checkpoint.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Supporting Adult Guidance v0.1.